

Monish Naresh Kumar

ROBOTICS SOFTWARE ENGINEER

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Nationality: Indian | Student Pass (Singapore) | **Available from October 2026**

PROFILE

A Robotics Software Engineer (graduating September 2026) specialising in ROS 2 autonomy, navigation, and Sim2Real development. My final-year project builds an autonomous Unitree Go2 care robot on ROS 2 with a single-owner motor architecture, LiDAR SLAM, obstacle veto, and an integrated perception and voice stack. I work across the full stack — from CoppeliaSim and Webots simulation to STM32 firmware and 3D-printed hardware — and I am proficient in Python, C/C++, and Git. Available from October 2026.

TECHNICAL SKILLS

Robotics & Autonomy: ROS 2 (Humble) · Nav2 · SLAM · finite-state machines · obstacle avoidance · person-following · CoppeliaSim · Webots · Unitree Go2 SDK · Sim2Real

Programming: Python (Advanced) · C, C++ · Lua · MATLAB · Git (Advanced) · pytest

Embedded & Hardware: STM32 (CubeIDE) · ESP32 · Arduino · XBee / UART · IR & ultrasonic · PCB design · Fusion 360 · 3D printing

Perception & AI: OpenCV · MediaPipe · YOLOv8 · ONNX · TensorFlow / Keras · LLM integration

Strengths: Systems integration · safety-first design · teamwork · clear communication · fast, independent learning

PROJECTS

Autonomous Robotics & ROS 2

Autonomous Medical Care Robot — Unitree Go2 — *ROS 2 Humble, Python, Nav2, SLAM · Final-Year Project (ongoing)*

- Architected a multi-process care-robot stack on ROS 2 in which a single motor arbiter — a priority FSM (ALERT > CHECK-IN > FALL > COMMAND > FOLLOW > PATROL) with a LiDAR obstacle veto — is the sole owner of robot motion.
- Implemented LiDAR SLAM mapping, idle patrol, obstacle avoidance, and person-following with distance hold.
- Integrated perception (YOLOv8 pose, fall detection, gesture/face recognition), a voice assistant, and ESP32 vitals into an authenticated live web dashboard; decision logic unit-tested with pytest.

Search-and-Rescue Mobile Robot Mission — *CoppeliaSim, Python, PyQt5*

- Programmed autonomous hazard-aware navigation with multi-reference path control and a custom PyQt5 operator interface.

Industrial Robotic Work-Cell Simulation (Dobot) — *CoppeliaSim, Lua*

- Scripted a Dobot pick-and-place cell (robot, sensors, spawner, conveyor) to model an efficient process flow.

Multi-Behaviour Robot Simulation — *Webots, Python*

- Implemented ten distinct autonomous behaviours across sensing, decision-making, and actuation.

Robotics Hardware & Embedded Systems

Line-Following & Obstacle-Avoidance Robot — Built from Scratch — *Fusion 360, 3D printing, Arduino / C++*

- Designed and 3D-printed the chassis, assembled the electronics end-to-end, and tuned line-following with obstacle avoidance.

Wireless Communication Between Two Microcontrollers — *STM32 (F401 / F411), XBee, UART, C*

- Built an interrupt-driven UART/XBee link exchanging ADC sensor data between two STM32 boards.

EDUCATION

Bachelor of Engineering in Robotics and Artificial Intelligence

PSB Academy, Singapore

Expected September 2026

VOLUNTEERING & LANGUAGES

Volunteer, PSB Academy (March & June 2026) — supported special events, program delivery, and participant feedback.

Languages: English (C1) · Tamil (B2) · Hindi (B1)